

Zen Nguyen

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TECHNICAL SKILLS

- **Languages:** Python, Go, Java, C, C++, TypeScript, JavaScript, SQL, HTML/CSS
- **AI & Data:** RAG Architecture, Vector Databases (FAISS/ChromaDB), LangChain, OpenAI API, Anthropic API, Pandas
- **Backend & Systems:** FastAPI, Node.js, Express, RESTful APIs, gRPC, TCP Sockets, Raft Consensus Algorithm
- **Infrastructure & DB:** Docker, PostgreSQL, MongoDB, Git, Linux (WSL)

TECHNICAL PROJECTS

TitanSwarm (Autonomous AI Co-Pilot) & TitanStore (Raft Database) Jan 2026 - Present
Personal Project · Python, Go, LangChain, FAISS

- Automated thousands of Software Engineering job postings by architecting an autonomous AI Co-Pilot in Python, enabling discovery, extraction, and parsing with advanced anti-bot evasion via concurrent JobSpy scrapers.
- Developed a zero-hallucination RAG engine using LangChain and OpenAI APIs, delivering uniquely tailored, ATS-optimized PDF resumes by leveraging only verified user data.
- Implemented the **Raft consensus protocol** and a custom Write-Ahead Log (WAL) in Go for TitanStore, a distributed memory bank, guaranteeing application state durability and fault tolerance across network failures.
- Designed and implemented a high-throughput **TCP Sockets** client API, enabling concurrent, real-time status updates to the Go database cluster, supporting seamless scaling to 100+ users.

SFU Course Tracker (sfucourseplanner.me) Nov 2025 - Jan 2026
Personal Project · Python, Docker, AWS, FastAPI

- Optimized deployments and reduced infrastructure costs by migrating a full-stack platform from Azure to AWS (EC2) and implementing **Docker** container orchestration.
- Engineered a scraping pipeline using Asyncio and HTTPX with semaphore-based rate limiting, increasing data throughput by 10x while respecting server constraints and adhering to **secure coding** practices.
- Designed and implemented a custom parser to tokenize and evaluate nested boolean prerequisite strings, converting unstructured text into a deterministic Abstract Syntax Tree (AST).
- Designed a normalized schema using SQLAlchemy with JSON-type columns to store recursive tree structures, optimizing read performance for complex queries.

Gridlock Casino (2D Arcade Engine) Feb 2026 - Present
Collaborative Project · Java, Swing, Maven, JUnit

- Managed project build automation and dependency resolution using Apache Maven to synchronize a 6-developer team, ensuring adherence to **secure product development** standards.
- Validated core engine logic through comprehensive **JUnit testing**, contributing to the overall quality and reliability of the game engine.

EDUCATION

Bachelor of Science, Computing Science May 2025 - Present
Simon Fraser University

- CGPA: 3.74 / 4.33
- Relevant Coursework: Discrete Math (A+), Computer Systems(A), Data Structures & Programming (B+), Linear Algebra(A-).
- Langara College | Associate of Science, Computer Science Jan 2024 - Apr 2025